

- Q.29 What is harmonic distortion? How does it arise in Class B operation?
- Q.30 Discuss the effect of diode characteristics on clamping circuits.
- Q.31 Explain the working of a Hartley oscillator with a neat circuit diagram.
- Q.32 What are the drawbacks in a electronic circuit not using proper feedback
- Q.33 Explain double tuned amplifiers
- Q.34 Draw the circuit for a push-pull amplifier and discuss its working
- Q.35 What is essential of an oscillator.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the operations of class B amplifier with neat circuit arrangement
- Q.37 Describe with neat circuit diagram and waveform of collector coupled astable multivibrator.
- Q.38 Explain the following:
- Diode clipping circuits.
 - Differentiator and integrator

No. of Printed Pages : 4
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180933/170933/120943

3rd Sem / Electrical Engineering, Power Station Engg. Elect & Eltx. Engg. Sub : Electronics – II

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is reactance?
- Resistance offered to AC current by resistor only
 - Resistance offered to DC current by resistor
 - Resistance offered to AC current by coil and capacitance
 - Resistance offered to DC current by coil and capacitance
- Q.2 What is the inductive reactance of a 40-mH coil at 18 Hmz?
- 4000Ω
 - 4522Ω
 - 4522H
 - 0
- Q.3 How will be the output voltage obtained for an ideal op-amp?
- Amplifies the difference between the two input voltages.
 - Amplifies individual voltages input voltage
 - Amplifies products of two input voltage
 - None of the mentioned
- Q.4 Which of the following is not an electronic device?
- A mobile
 - A computer
 - A magnifying glass
 - A keyboard

- Q.5 What are oscillators?
 a) Switching circuits b) Converts DC to AC
 c) Converts AC to DC d) Filter circuits
- Q.6 Which of the application's filter used for?
 a) Reducing ripples
 b) Increasing ripples
 c) Increasing phase change
 d) Increasing amplitude
- Q.7 Which among the following is not a special case of voltage shut feedback amplifier?
 a) Voltage followers
 b) Current to voltage connector
 c) Inverter
 d) None of the mentioned
- Q.8 What is the power dissipation of power transistor?
 a) @0.5 W b) £ 0.5W
 c) >0.5W d) ¹ 0.5W
- Q.9 Which amplifier provides twice output swing as that of LM380 amplifier?
 a) Hybrid power amplifier
 b) Bridge power audio amplifier
 c) Monolithic power audio amplifier
 d) Dual power amplifier
- Q.10 Write the formula for closed loop voltage gain of inverting amplifier with feedback using open loop voltage gain and gain of feedback circuit.
 a) $A_F = A/(1+AB)$
 b) $A_F = -A(1+AB)$
 c) $A_F = -B(1+AB)$
 d) None of the mentioned

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is meant by feedback?
 Q.12 What type of feedback is used in oscillator?
 Q.13 CMRR means _____.
 Q.14 Voltage gain is _____.
 Q.15 The generator capacitance is dependent on the design of the generator. (True / False)
 Q.16 Main Parameter of Power Amplifier is _____ Efficiency.
 Q.17 Unit of power is _____.
 Q.18 CVT means _____.
 Q.19 Can a voltmeter measure e.m.f.? Yes/No
 Q.20 What component is used as load?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write short note on CVT.
 Q.22 Briefly explain the design of transistor switch.
 Q.23 Explain the Concept of Negative Feedback and its Effects on Amplifier Performance.
 Q.24 Explain operations of 555 IC timer with neat diagram.
 Q.25 Write application of tuned voltage amplifiers.
 Q.26 Draw a circuit for class C amplifier and discuss its working?
 Q.27 Write application of wave shaping circuits.
 Q.28 Explain emitter follower and write any two applications.